

C and C++ Programming Topics

C Programming Topics

Category	Topics Covered
Overview of C	History of C, Features of C, Basic Structure of C Program, Sample Structure of C Program
Understanding Memory Map	Memory Organization, Memory Organization in DOS Case
Constants, Variables & Data Types	Rules for Declaring Variables, Types of Variables (Local, Global, Static, Register), Keywords & Constants, Data Types, Representation in Memory, Overflow/Underflow
Operators & Expressions	Types of Operators, Binary/Unary/Ternary Operators, Bitwise Operators, Operator Precedence, Type Conversion, Logical & Conditional Expressions
Control Structures - Branching	if Statement, if-else, Nested if-else, switch-case, goto, Ternary Operator
Control Structures - Looping	while, for, do-while, Jump in Loop
Arrays & Strings	Definition, Types (1D, 2D, Multi D), Declaration & Initialization, Reading/Writing Strings, String Handling Functions, Arrays & Pointers
Handling Pointers	Rules, Pointer Arithmetic, Array of Pointers, Passing Address through Pointers, Pointers to Functions
Functions & Variables	Variable Scope, Storage Classes, Declaration & Definition Scope, Initialization, Creation and Derived Data
Functions	Why Functions, Types, User-defined Functions, Functional Program, Function Parameters, Recursion, Arguments & Return Types
Structures & Unions	Defining Structures, Initialization, Nested Structures, Array of Structures, Pointers with Structures, Union & Its Properties
File Handling in C	What is File, Why Use File, Types of Files, Reading & Opening File, File Handling Operations

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C++ Programming Topics

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Introduction to OOPs	Modular vs OOPs, Features of OOPs, Benefits
Classes & Objects	Abstract Data Types, Class Declaration, Class Definition, Creating Object, Defining Class & Accessing Members
Functions & Variables	Scope, Storage Classes, Declaration & Definition Scope, Initialization, Creation and Derived Data
Using Constructors	Overview, Multiple Constructors, Initialization Lists, Base & Derived Class Constructors, Object Creation, Member Function Definition, Constructor/Destructor
Inheritance	Overview of Inheritance, Defining Base and Derived Classes
Polymorphism	Overview, Function Overloading, Constructor Overloading, Operator Overloading, Defining Polymorphism
Exception Handling	Exception Overview, try/catch/throw, Standard Exception Handler, Exception Handling with Inheritance
Templates	Templates Overview, Standard Template Method, Creating & Using Templates, Library Containers, File Handling Required